Review plan · for the teacher

Exploring data: unit check

Data science · Level 2 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Histograms and shapes
- Spread and outliers
- Scatter and correlation
- A routine for exploring

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet for its answers.
2. Quiz (20 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask a team to explain it with an example from the unit.
4. Wrap-up (8 min): Work through the question most teams got wrong, together on the board.

Discussion

- Which idea will you use first?
- Which table will you explore?
- Which claim will you question?

Exit ticket

- One finding I will look for this week, and where.

Try it at school tomorrow

Explore one school table with a class using the five questions, and share three findings with numbers and sources.

Quiz sheet · for each team

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Team _____ Date _____

1 **What does each bar of a histogram show?**

◯ One pupil's value ◯ How many values fall in its bin ◯ The mean

2 **In data skewed right, the median is usually above the mean.**

◯ True ◯ False

3 **Match each term to its meaning.**

Standard deviation	• • • •	Spread of the middle half
IQR	• • • •	A value far from the rest
Outlier	• • • •	Typical distance from the mean

4 **Complete the rule.**

Q1 is 20 and Q3 is 28, so the upper fence is _____ .

Word bank: 36 · 40

5 **A journey time of 300 minutes appears. What do you do first?**

◯ Delete it ◯ Check whether it is a typo or true ◯ Report it as the average

6 **Which describe a correlation of about −0.8? Choose all that apply.**

☐ Dots fall from left to right ☐ One causes the other ☐ The pattern is fairly strong

7 **Ice cream sales and drownings rise together, so ice cream causes drowning.**

◯ True ◯ False

8 **Put the exploring routine in order.**

☐ Gaps and oddities

☐ Centre and spread

☐ Groups and relationships

☐ What is in it?

☐ Shape

9 **Which is the best finding?**

◯ Library loans fall by half in exam weeks (library log, term 2) ◯ Pupils don't like reading

◯ The library is useful

10 **Very wide bins can hide the shape of a distribution.**

◯ True ◯ False

Answers · for the teacher

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1 **How many values fall in its bin**

A histogram counts values in equal bins.

2 **False**

The tail of large values pulls the mean above the median.

3 **Standard deviation — Typical distance from the mean**
IQR — Spread of the middle half
Outlier — A value far from the rest

These three describe spread and what doesn't fit.

4 **Q1 is 20 and Q3 is 28, so the upper fence is 40.**

IQR = 8, and 28 + 1.5 × 8 = 40.

5 **Check whether it is a typo or true**

Outliers are questions: check them before deciding.

6 **Dots fall from left to right; The pattern is fairly strong**

Negative means falling and 0.8 is strong; neither says anything about cause.

7 **False**

Hot weather drives both. Correlation is not causation.

8 **Top to bottom: 4, 3, 5, 1, 2**

From the whole table, to each column, to problems, to comparisons.

9 **Library loans fall by half in exam weeks (library log, term 2)**

It is specific, has a number and names its source.

10 **True**

Two or three fat bars lump everything together.